

Supplementary Information: Data Quality Assurance Analysis Keheala Study 2

Contents

1	Patient Characteristics	2
2	Data Quality Assurance: Outcome Discrepancies	3
2.1	Crosstab of TIBU vs. Paper Registry Outcomes	3
2.2	Error Rates by Clinic Characteristics	4
2.3	Error Rates by Patient Characteristics	5
2.4	Age of Record by Error Type	5
2.5	Error Rates Over Time	5
2.6	Error Rates by Age of Record	6

Overview

This supplementary document presents descriptive statistics and data quality assurance (DQA) analyses conducted as part of the Keheala Study 2 randomized controlled trial, which evaluated a digital adherence support platform for tuberculosis (TB) patients in Kenya. The DQA assesses recording errors in TIBU, Kenya’s national electronic TB reporting system, by treating a contemporaneous paper registry audit as the ground truth. The TIBU dataset available for this study covers patients registered between April 2018 and December 2019 ($N = 119,653$ nationwide). A paper registry audit was conducted across study clinics between November 2018 and September 2019, yielding 3,645 audited records, of which 3,370 could be matched to a corresponding TIBU record and form the basis of the DQA. Tables 1a and 1b characterize the study population across three nested cohorts: all patients registered in TIBU during the study period, patients at study clinics in TIBU (the MITT sample), and the subset of study-clinic patients whose records were audited in the DQA.

The core of the DQA compares treatment outcomes recorded in TIBU against outcomes recorded in the paper registry for each matched patient. Patients recorded as transferred out in TIBU—including those assigned the MT4 code, used when drug resistance is discovered mid-treatment and the patient is referred to a drug-resistant TB program—are excluded from error classification, as a transfer is not an auditable final outcome. Records where the paper registry has no outcome are also excluded, as the DQA requires ground truth. Among the remaining patients, three types of discrepancy are distinguished. Type I errors occur when TIBU records a successful outcome (treatment completion or cure) but the paper registry records a failure (death, treatment failure, or loss to follow-up), representing a false positive in the digital record. Type II errors are the reverse: TIBU records a failure while the paper registry records success, representing a false negative. False-missing discrepancies arise when TIBU contains no outcome but the paper registry does, representing a gap in the digital record. The cross-tabulation in Table 2 maps the full distribution of these discrepancies across outcome categories; subsequent tables stratify error rates by clinic characteristics (province, urbanicity, and size) and patient characteristics (sex, age, disease type, and HIV status); and the smoothed time-series figures trace how error rates vary with the age of the patient record and across the study period.

1 Patient Characteristics

Tables 1a and 1b describe the study population across three cohorts: all patients registered in the TIBU electronic system during the study period, patients at study clinics recorded in TIBU (MITT sample), and patients at study clinics whose records were audited in the paper registry DQA.

Table 1a: Individual and disease characteristics across TIBU and paper registry cohorts.

Characteristic	Digital				Paper	
	All		Study clinics		Study clinics	
	(N=119,653)		(N=15,487)		(N=3,370)	
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)
<i>Individual characteristics</i>						
Male	75,597	(63.2%)	9,850	(63.6%)	2,104	(64.5%)
Mean age (years)	36.1		34.4		32.7	
<i>Disease characteristics</i>						
Bacteriologically confirmed	62,149	(51.9%)	11,663	(75.3%)	2,295	(70.4%)
Drug resistant	1,343	(1.1%)	2	(0.0%)	3	(0.1%)
Extrapulmonary TB	17,558	(14.7%)	1,834	(11.8%)	369	(11.3%)
HIV coinfection	32,077	(26.8%)	4,118	(26.8%)	746	(23.3%)
Retreatment	8,351	(7.0%)	1,308	(8.7%)	297	(9.3%)

Table 1b: Treatment outcomes across TIBU and paper registry cohorts.

Characteristic	Digital				Paper	
	All		Study clinics		Study clinics	
	(N=119,653)		(N=15,487)		(N=3,370)	
	<i>N</i>	(%)	<i>N</i>	(%)	<i>N</i>	(%)
<i>Outcomes</i>						
Cured	25,399	(21.2%)	7,008	(45.3%)	1,148	(34.1%)
Treatment completed	31,719	(26.5%)	6,851	(44.2%)	1,520	(45.1%)
Died	4,911	(4.1%)	787	(5.1%)	186	(5.5%)
Treatment failed	382	(0.3%)	129	(0.8%)	37	(1.1%)
Lost to follow-up	3,594	(3.0%)	712	(4.6%)	133	(3.9%)
Not completed	47,184	(39.4%)	0	(0.0%)	87	(2.6%)
Transfer out	6,382	(5.3%)	0	(0.0%)	139	(4.1%)
Not TB	0	(0.0%)	0	(0.0%)	0	(0.0%)
Missing	82	(0.1%)	0	(0.0%)	120	(3.6%)

2 Data Quality Assurance: Outcome Discrepancies

2.1 Crosstab of TIBU vs. Paper Registry Outcomes

Table 2 shows the full cross-tabulation of treatment outcomes recorded in TIBU against outcomes recorded in the paper registry for all matched patients ($N = 3,370$).

Table 2: Cross-tabulation of treatment outcomes in TIBU vs. paper registry.

Outcome in TIBU	Treatment Outcome in Paper Registry							Total
	Blank <i>N</i> (%)	C <i>N</i> (%)	D <i>N</i> (%)	F <i>N</i> (%)	LTFU <i>N</i> (%)	TC <i>N</i> (%)		
Blank	80 (66.7%)	22 (18.3%)	2 (1.7%)	2 (1.7%)	2 (1.7%)	12 (10.0%)	120	
C	45 (3.9%)	1,032 (89.9%)	1 (0.1%)	3 (0.3%)	7 (0.6%)	60 (5.2%)	1,148	
D	4 (2.2%)	3 (1.6%)	170 (91.4%)	0 (0.0%)	1 (0.5%)	8 (4.3%)	186	
F	0 (0.0%)	1 (2.7%)	1 (2.7%)	28 (75.7%)	5 (13.5%)	2 (5.4%)	37	
LTFU	3 (2.3%)	2 (1.5%)	1 (0.8%)	0 (0.0%)	121 (91.0%)	6 (4.5%)	133	
NC	10 (11.5%)	34 (39.1%)	6 (6.9%)	0 (0.0%)	4 (4.6%)	32 (36.8%)	87	
TC	34 (2.2%)	81 (5.3%)	8 (0.5%)	0 (0.0%)	13 (0.9%)	1,384 (91.1%)	1,520	
TO	6 (4.3%)	46 (33.1%)	5 (3.6%)	4 (2.9%)	14 (10.1%)	63 (45.3%)	139	
Total	182	1,221	194	37	167	1,567	3,370	

Note: Percentages are row shares.

Red = Type I error (TIBU: success; paper: failure).

Green = Type II error (TIBU: failure; paper: success).

Blue = Incomplete data (TIBU: no outcome; paper: outcome present).

2.2 Error Rates by Clinic Characteristics

Table 3 reports false-negative, false-positive, and false-missing rates stratified by province, urbanicity, and clinic size.

Table 3: DQA error rates by province, urbanicity, and clinic size.

	<i>N</i> (%)	False positive <i>N</i> (%)	False negative <i>N</i> (%)	Incomplete data <i>N</i> (%)
<i>By province</i>				
Central	25 (0.8%)	0 (0.0%)	0 (0.0%)	12 (48.0%)
Coast	713 (23.3%)	7 (1.0%)	2 (0.3%)	12 (1.7%)
Eastern North	2 (0.1%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Eastern South	264 (8.6%)	4 (1.5%)	2 (0.8%)	22 (8.3%)
Nairobi	376 (12.3%)	6 (1.6%)	5 (1.3%)	3 (0.8%)
North Eastern	745 (24.4%)	9 (1.2%)	3 (0.4%)	2 (0.3%)
Nyanza North	390 (12.8%)	2 (0.5%)	7 (1.8%)	3 (0.8%)
Nyanza South	43 (1.4%)	1 (2.3%)	0 (0.0%)	0 (0.0%)
Rift Valley North	41 (1.3%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Rift Valley South	391 (12.8%)	2 (0.5%)	2 (0.5%)	29 (7.4%)
Western	32 (1.0%)	1 (3.1%)	1 (3.1%)	1 (3.1%)
<i>By urban/rural</i>				
Urban	965 (31.6%)	11 (1.1%)	6 (0.6%)	18 (1.9%)
Rural	2,053 (67.2%)	21 (1.0%)	16 (0.8%)	66 (3.2%)
<i>By clinic size</i>				
Large clinic	2,361 (77.3%)	30 (1.3%)	16 (0.7%)	53 (2.2%)
Small clinic	693 (22.7%)	2 (0.3%)	6 (0.9%)	63 (9.1%)
All	3,054 (100.0%)	32 (1.0%)	22 (0.7%)	116 (3.8%)

Note: Urban/rural classification is based on clinic records. Large clinics are those with at least 2 patients registered in TIBU (median); small clinics are those with fewer.

2.3 Error Rates by Patient Characteristics

Table 4 reports false-negative, false-positive, and false-missing rates broken down by individual characteristics and disease characteristics.

Table 4: DQA error rates by individual and disease characteristics.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Incomplete data <i>N</i>	(%)
<i>Individual characteristics</i>								
Male (%)	1,936	(63.4%)	20	(1.0%)	14	(0.7%)	68	(3.5%)
<i>Age group</i>								
<15	433	(14.2%)	1	(0.2%)	3	(0.7%)	5	(1.2%)
15–34	1,379	(45.2%)	18	(1.3%)	12	(0.9%)	36	(2.6%)
35–54	897	(29.4%)	7	(0.8%)	5	(0.6%)	29	(3.2%)
55+	306	(10.0%)	6	(2.0%)	2	(0.7%)	8	(2.6%)
<i>Disease characteristics</i>								
Bacteriologically confirmed	2,108	(69.0%)	26	(1.2%)	15	(0.7%)	61	(2.9%)
Extrapulmonary	340	(11.1%)	2	(0.6%)	0	(0.0%)	9	(2.6%)
Retreatment	281	(9.2%)	5	(1.8%)	1	(0.4%)	5	(1.8%)
HIV positive	695	(22.8%)	7	(1.0%)	9	(1.3%)	21	(3.0%)
All	3,054	(100.0%)	32	(1.0%)	22	(0.7%)	116	(3.8%)

2.4 Age of Record by Error Type

Table 5 reports two measures of record age at the TIBU download snapshot, stratified by error type: (i) record age, measured from patient registration to the snapshot, and (ii) reporting lag, measured from the TIBU outcome date to the snapshot.

Table 5: Record age and reporting lag (days) at the TIBU snapshot, by error category.

Error type	N (%)		Record age (days)			Reporting lag (days)		
			n	Mean	Median	n	Mean	Median
False positive (Type I)	32	(1.0%)	32	1013	980	32	866	801
False negative (Type II)	22	(0.7%)	22	1047	1040	22	981	1026
False missing	116	(3.8%)	116	928	844	114	758	696
No error	2,884	(94.4%)	2,884	1030	1004	2,884	848	850

Note: The TIBU snapshot is taken to be 1 January 2022, based on the latest treatment-outcome dates in the extract clustering in late December 2021. Record age is measured from registration to snapshot; reporting lag is measured from TIBU outcome date to snapshot. Column n gives the number of records with a valid date; N counts all records in the error category.

2.5 Error Rates Over Time

The TIBU dataset used for these analyses reflects a single final extract, with the latest treatment-outcome dates clustering in late December 2021; we treat the snapshot as having occurred on 1 January 2022. Figure 2 plots DQA error rates against outcome date, estimated via locally-weighted regression (lowess); shaded bands show percentile bootstrap 95% confidence intervals (500 resamples). The outcome date is TIBU’s when available and the paper registry’s otherwise (used for false-missing records, where TIBU is blank).

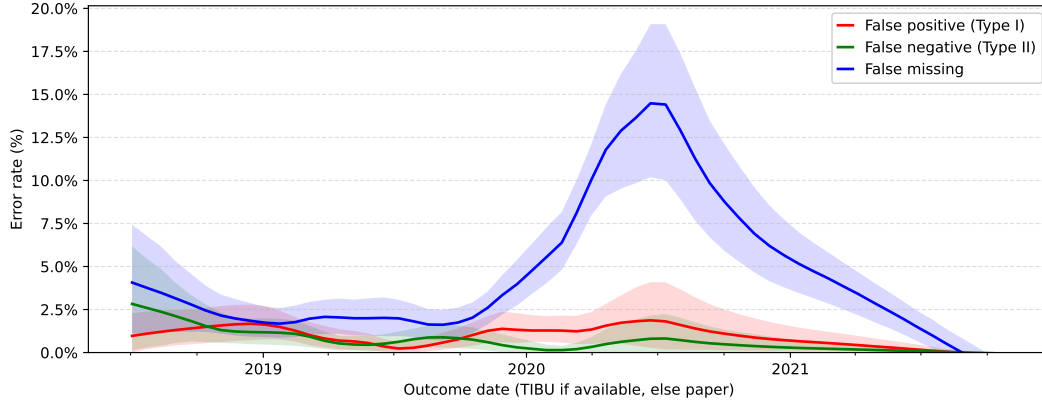


Figure 2: DQA error rates over time by outcome date (TIBU when available, paper otherwise). Lowess smooth with percentile bootstrap 95% CI bands.

The dominant feature of the figure is a pronounced peak in the false-missing rate (to roughly 15%) for records with outcome dates in 2020 Q1–Q2, coinciding with the onset of the COVID-19 pandemic response in Kenya. Of the 116 false-missing records in the classifiable sample, 63 fall in these two quarters, and 50 are concentrated at just two clinics whose overall false-missing rates (12.7% and 15.4%) are an order of magnitude above the study-wide rate of 3.8%. At these clinics, paper-registry entries continued through the pandemic but TIBU outcomes were largely not recorded. Treating the TIBU NC (“not completed”) code as a placeholder rather than a medical outcome rolls the affected records into the false-missing category; this accounts for 76 of the 116 (the remaining 40 have TIBU entirely blank). Type I and Type II error rates are both low and roughly flat across the study period.

2.6 Error Rates by Age of Record

Figures 3–5 re-express the same data against record-level age axes. Figure 3 plots error rates against record age at snapshot (days from registration to 1 January 2022); Figure 4 plots them against reporting lag (days from outcome date to snapshot, with the axis inverted so that calendar time reads left-to-right like Figure 2); Figure 5 compares the empirical distributions of reporting lag across error categories. The same COVID-era false-missing peak is visible as a left-edge spike in Figure 3 (records registered in late 2019, whose treatments completed in 2020) and as a mid-2020 spike in Figure 4; the ECDF in Figure 5 shows the false-missing distribution shifted noticeably toward shorter reporting lags, which is the same COVID-era concentration viewed as a cumulative distribution.

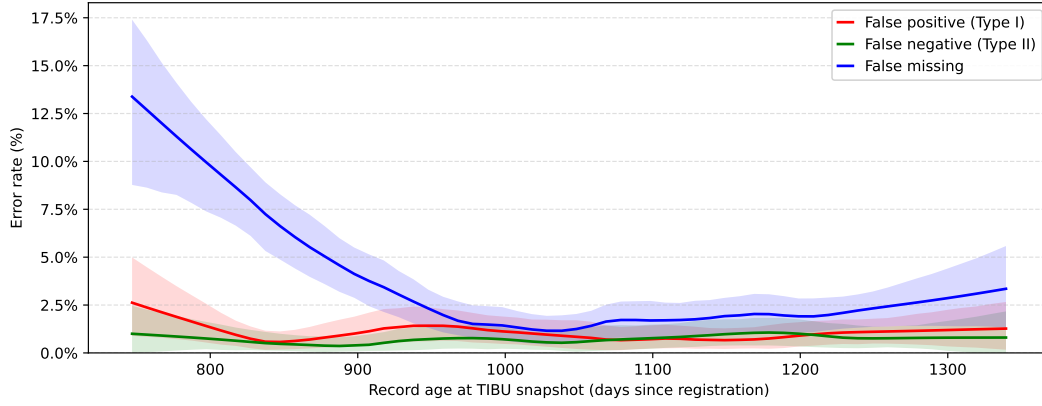


Figure 3: DQA error rates by record age at TIBU snapshot (days since registration). Lowess smooth with percentile bootstrap 95% CI bands.

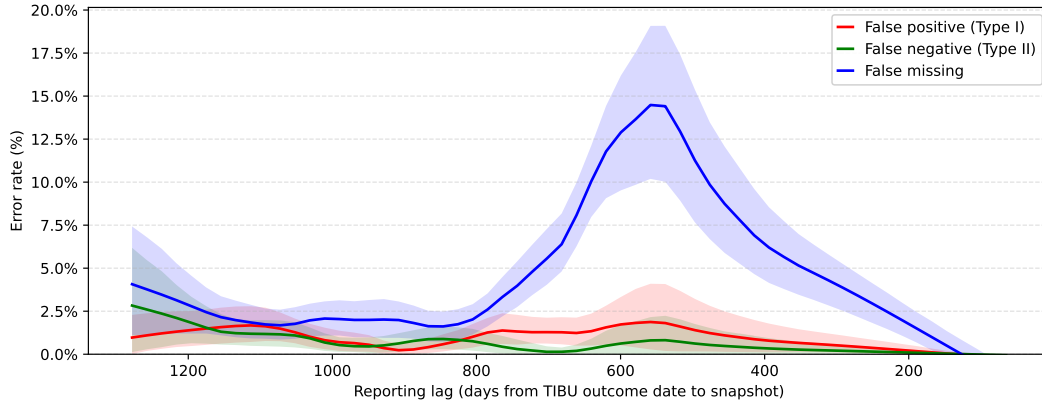


Figure 4: DQA error rates by reporting lag (days from the outcome date to the TIBU snapshot); x-axis inverted so older outcomes are on the left and recent outcomes on the right. Lowess smooth with percentile bootstrap 95% CI bands. For false-missing records (TIBU blank) the paper-registry outcome date is used in place of the TIBU outcome date.

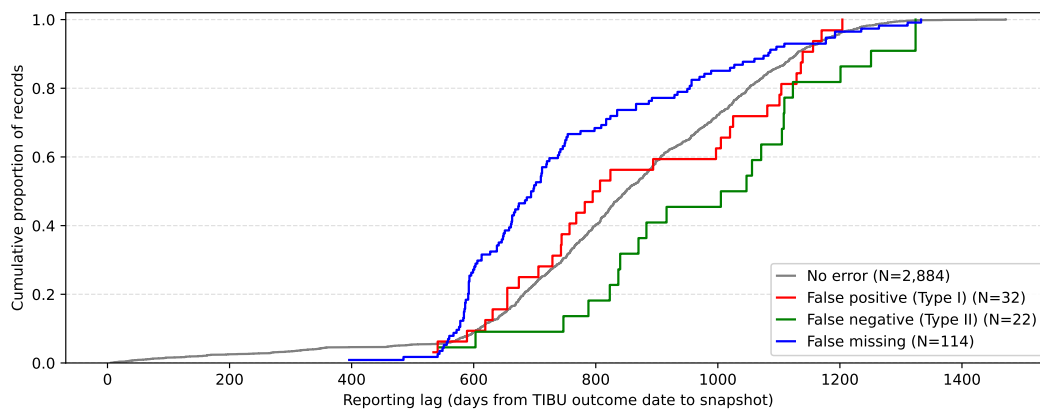


Figure 5: Empirical cumulative distribution of reporting lag, by error category. Separation from the no-error curve indicates where on the age axis each error type concentrates.